

KORVANE COLD CHAIN — CALIBRATION CERTIFICATE

Form KC-CAL-T, issue 4, valid from 01.03.2024 (replaces issue 3 of 2022).
To be completed in ink or by the KC-Service Tool export. Fields marked with an asterisk are mandatory; a certificate with an empty mandatory field is void and must be reissued with a new number.

Certificate no.*: KCC-2024-_____ (five digits, assigned in running order by Bay 3; the numbers are never reused, even if a certificate is cancelled)

Issued by: Korvane Cold Chain Sp. z o.o., Calibration Laboratory, Bay 3, building GDY-1, ul. Łużycka 14, 81-537 Gdynia

1. Device under test

Field	Entry
Device type*	KC-Tag 4 (KT4-R2-868) / KC-Tag 4 R1 — delete one
Serial number*	KT4-_____ - _____
Firmware version at the start*	_____
Customer / order reference	_____
Date of calibration*	_____._____.20____
Technician*	_____

2. Reference equipment

Item	Entry
Dry-block calibrator	TB-120, asset no. _____
Reference probe	RX-400, serial 88213; certificate ACC-LAB-2024-0113 of 12 February 2024
Humidity chamber	HC-2 with LiCl (11.3 % RH) and NaCl (75.3 % RH) at +25 °C
Ambient conditions	_____ °C / _____ % RH

The reference probe is traceable to the national standard through the accredited laboratory Thermetrix Kalibracja Sp. z o.o. in Gdańsk (the certificate named above). The certificate is valid for 12 months and the next calibration of the probe is due in February 2025.

3. Results — temperature

Offset = reading of the device minus reference. All values in °C.

Point	Reference (°C)	Offset as found	Offset after adjustment	Limit
1	+40.0	_____	_____	±0.25 as

				found; ± 0.10 after adjustment
2	0.0	_____	_____	± 0.25 as found; ± 0.10 after adjustment
3	-25.0	_____	_____	± 0.25 as found; ± 0.10 after adjustment

For KC-Tag 4 R1 devices the as-found limit is ± 0.40 °C.

4. Results — humidity

Point	Reference (% RH)	Offset as found (% RH)	Offset after adjustment (% RH)	Limit
1	11.3	_____	_____	± 2.0
2	75.3	_____	_____	± 2.0

Humidity points are adjusted only when the customer needs humidity (see the service manual KC-SM-4, chapter 7.4).

5. Statement

☐ As found: within limits, no adjustment made. ☐ Adjusted: offsets written to the device, re-measured within limits. ☐ Rejected: see rejection report (sensor board replaced, calibration repeated).

The expanded uncertainty of measurement for the temperature points is ± 0.08 °C (coverage factor $k = 2$, confidence level approximately 95 %). The statement of conformity is made with the simple acceptance rule: no guard band is applied.

Next calibration due*: _____.____.20__ (12 months after the date of calibration). The device displays the error E12 from 13 months after this date.

6. Signatures

Technician: _____ Date: _____ Checked by (second person, mandatory for pharma customers): _____ Date: _____

Notes for the issuer (not printed on the customer copy):

1. The original is filed in the grey cabinet, Bay 3, shelf 2; a PDF copy is attached to the device record in Tidewalk within one working day.
2. Retention period 7 years.
3. Customers under a pharma contract receive the certificate within 24 hours after the calibration, others within 5 working days.
4. Forms printed before 01.03.2024 (issue 3) must be destroyed; the old form had only a two-point temperature check, which is not accepted any more.
5. Issue 4 was approved by Benedikta Aaltonen on 26 February 2024.

Page 1 of 1. This certificate shall not be reproduced except in full without the written approval of the laboratory.